



Why Monitoring AI Systems Matters

Ensuring reliability, performance, and trust in production AI

The Monitoring Mindset



AI ≠ traditional apps

Failures are subtle and often silent, making detection challenging without proper visibility



Visibility into black-box models

Monitoring provides crucial insights into otherwise opaque AI systems



Detect issues early

Identify drift, errors, and outages **before** users or regulators do

Why AI Needs Special Monitoring

Non-determinism

Same input may produce different outputs across runs

Data dependency

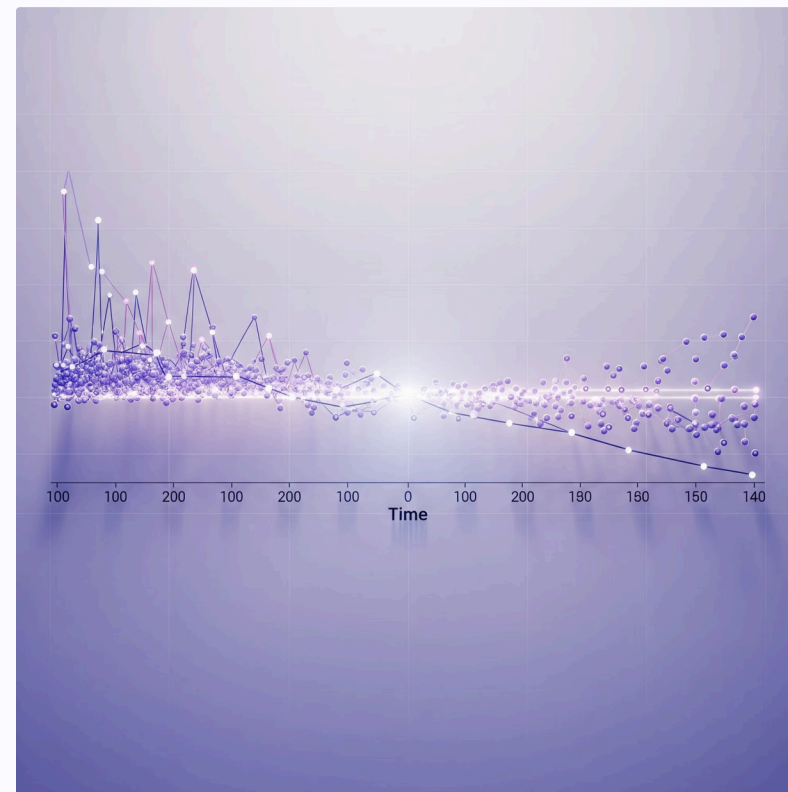
Quality of input directly drives quality of output

Model drift

Accuracy degrades over time as the world changes

Infrastructure sensitivity

GPUs, memory, and queues significantly impact latency



Dimensions to Monitor



Infrastructure metrics

- CPU/GPU utilization
- Memory consumption
- Network throughput



Model performance

- Accuracy and F1 scores
- Perplexity measurements
- Drift detection metrics



Operational metrics

- Latency (p95/p99)
- Request throughput
- Error and failure rates



Business KPIs

- Conversion rates
- Fraud detection accuracy
- Customer satisfaction

Failure Modes Without Monitoring

Silent performance drops

Gradual degradation goes unnoticed until users lose trust

Infrastructure bottlenecks

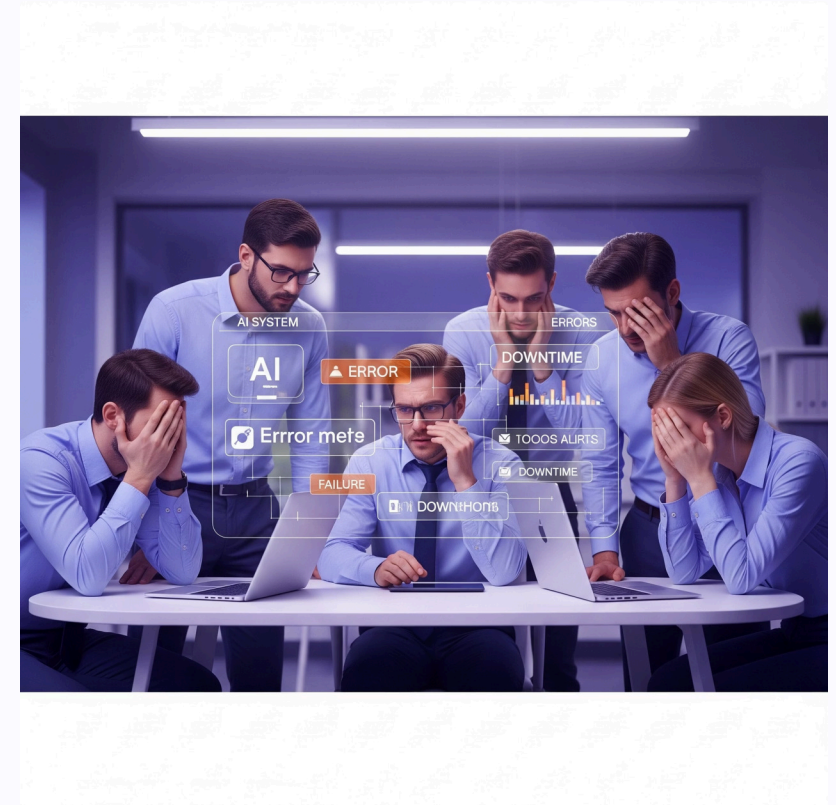
Systems crash during peak load events without warning

Data pipeline errors

Garbage in, garbage out - corrupted inputs produce invalid predictions

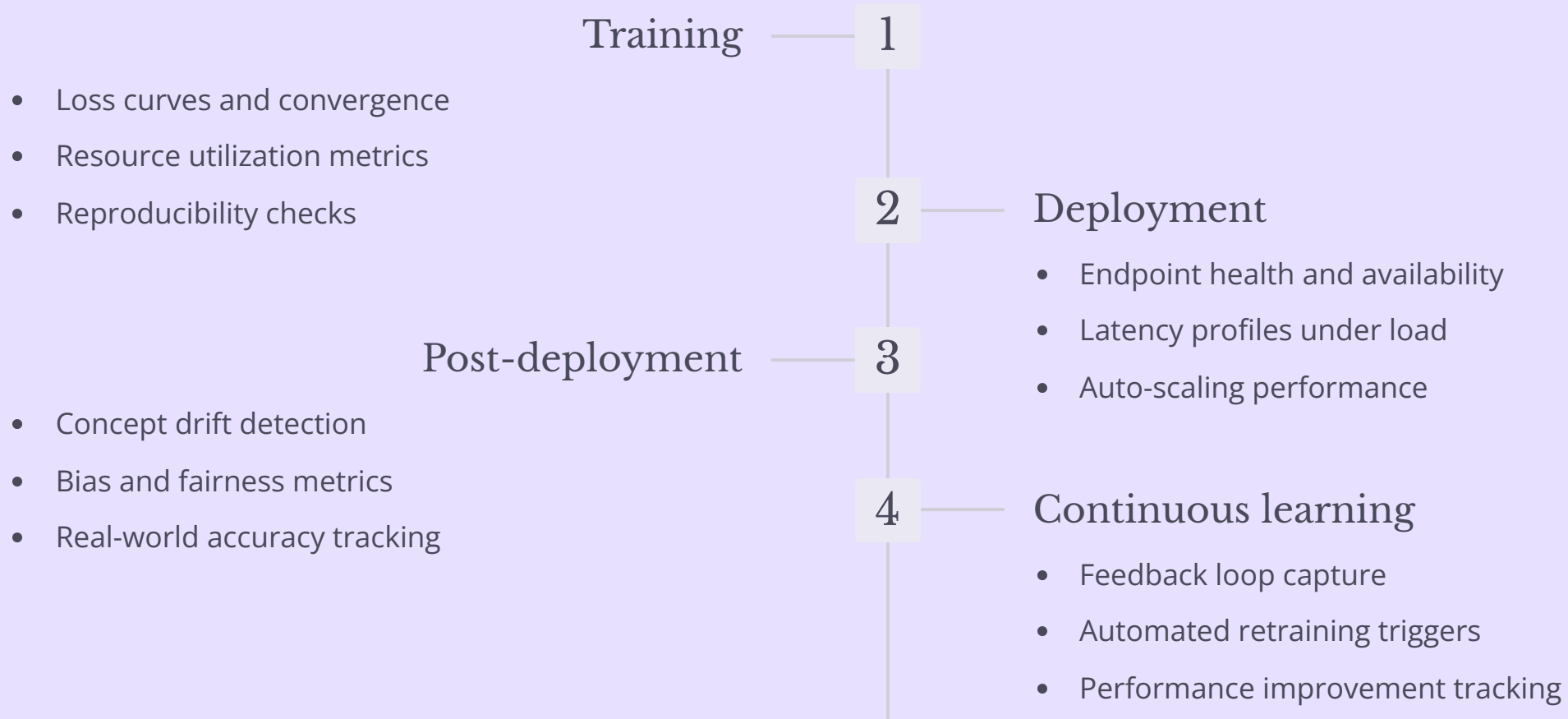
Compliance violations

Uncaught bias or fairness issues lead to legal and financial risk



Without proper monitoring, issues cascade into major failures

Monitoring Across the Lifecycle



Each phase requires specific monitoring approaches to maintain system health

Tools & Ecosystem

Infrastructure

Prometheus, Grafana, DCGM

Monitor hardware health, resource usage, and GPU metrics

Model Operations

MLflow, Weights & Biases, EvidentlyAI

Track experiments, model lineage, and detect production drift

Observability

OpenTelemetry, ELK stack, Datadog

Collect and analyze traces, logs, and metrics across systems

Business Layer

Custom dashboards integrated with business KPIs

Connect technical metrics to business outcomes



Modern AI systems require an integrated monitoring stack

Best Practices



Define clear SLOs

Set specific targets for latency, accuracy, and uptime

Example: **p99 prediction latency < 200ms**



Monitor distribution, not averages

Track p50/p95/p99 metrics to catch tail issues

Averages hide critical performance problems



Combine signal types

Correlate infrastructure, model, and business metrics

Look for relationships between technical and business impact



Build alerts and playbooks

Actionable alerts with clear response procedures

Reduce mean time to resolution (MTTR)

Key Takeaways

Monitoring is not optional

It's core to AI reliability, performance, and trust

Comprehensive coverage

Infrastructure, model, operational, and business layers

Protection and optimization

Safeguards user trust, ensures compliance, optimizes costs



Continuous monitoring is the foundation of MLOps maturity